GLYCEMIC TWIN

Two-stream fusion / physiology-informed prediction / physician review

SYNTHETIC RESEARCH PoC

No clinical validation or approval

01 Static EHR

40 synthetic patient profiles
HbA1c, HOMA-IR, C-peptide
BMI, eGFR, TCF7L2, medication
Baseline record precedes telemetry

02 Wearable stream

53,760 packets; 15-minute cadence
CGM, HR / HRV, steps, energy
Meals + last completed night's sleep
14 days; India scenario; seed 2026

03 Safety quarantine

Validate status + CGM 40-400 mg/dL

Single missing packet in engine:
past-only extrapolation; slope bound
No recursive repair or future access

Repeated missing / corrupt packets:
lock the affected lookback window

Dashboard: every invalid current
packet locks scores + scenario UI
Historical repair remains flagged

04 Metabolic fingerprint

Join by patient_id at timestamp t
Strict history: [t - 180 min, t]

26 inputs / 29 encoded columns
Glucose level, trend, variability
GI-weighted absorption + activity
Autonomic stress + cyclic time

EHR x glycemic load
BMI x muscular clearance
Sleep x autonomic stress

Scaler learns from fitting patients only

05 Multi-split ML engine

Patient partitions: 60:40 / 70:30 /
80:20 / 90:10; 4 archetypes held out

A: logistic classifier + ridge trajectory
B: histogram gradient boosting
Sigmoid probability calibration
Model selection uses inner validation

Two outputs from the exported bundle:
P(any future CGM > 180 mg/dL)
8 glucose values: +15 to +120 min
Future labels use raw valid CGM only

06 Physiology-informed interactions

Glycemic load = carbs(2h) x GI/60 x HOMA-IR/3 x (1 + 0.07 x alleles)
Clearance = steps(1h)/100 x 25/BMI
Cortisol proxy = (100 - sleep efficiency) x HR/HRV
Exercise and stress are distinguished using activity + autonomic signals

07 Physician CDS + scenario sandbox

Held-out replay: baseline forecast, calibration RMSE band, adjustable alert threshold
Walk / meal / GI / breathing edits recompute features and call the fitted models
Exact grouped Shapley contributions; reviewable clinician note; timestamped audit
Scenarios are feature perturbations: intervention benefit is not causally established

PRIMARY HOLDOUT | AUROC 0.993 | Recall 91.1% | Trajectory RMSE 10.16 mg/dL | New-crossing recall 51.9%

80:20 synthetic cohort; classifier threshold 0.50. Metrics are read from the saved evaluation. No diagnostic, dosing or causal-treatment claim.